PANEL LABELING

CDP - 6 - N - 1 - L - 1

- SEQUENCE NUMBER 1, 2, 3, ...
- TYPE
  - L = LIGHTING
  - P = POWER
  - M = MECHANICAL
  - D = DISTRIBUTION
- LOCATION/FLOOR
  - 0 = LEVEL 0
  - 1 = LEVEL 1
  - 2 = LEVEL 2
  - ...
  - R = ROOF
- SYSTEM
  - N = NORMAL
  - E = LIFE SAFETY
  - S = STANDBY
- VOLTAGE
  - 6 = 600Y/347V
  - 2 = 208Y/120V
- ABBREVIATIONS:
  - MDP = MAIN DISTRIBUTION PANEL
  - CDP = CENTRAL DISTRIBUTION PANEL
  - SPL = SPLITTER
  - MCC = MOTOR CONTROL CENTRE

kA = CALCULATED SHORT CIRCUIT CURRENT

TRANSFORMER LABELING

TX-N - 0 - 1

- SEQUENCE NUMBER 1, 2, 3, ...
- LOCATION/FLOOR
  - 0 = LEVEL 0
  - 1 = LEVEL 1
  - 2 = LEVEL 2
  - R = ROOF
- SYSTEM
  - N = NORMAL
  - E = LIFE SAFETY
  - S= STANDBY

600V - 208V TRANSFORMER BREAKER, FEEDER AND CONDUIT SIZING

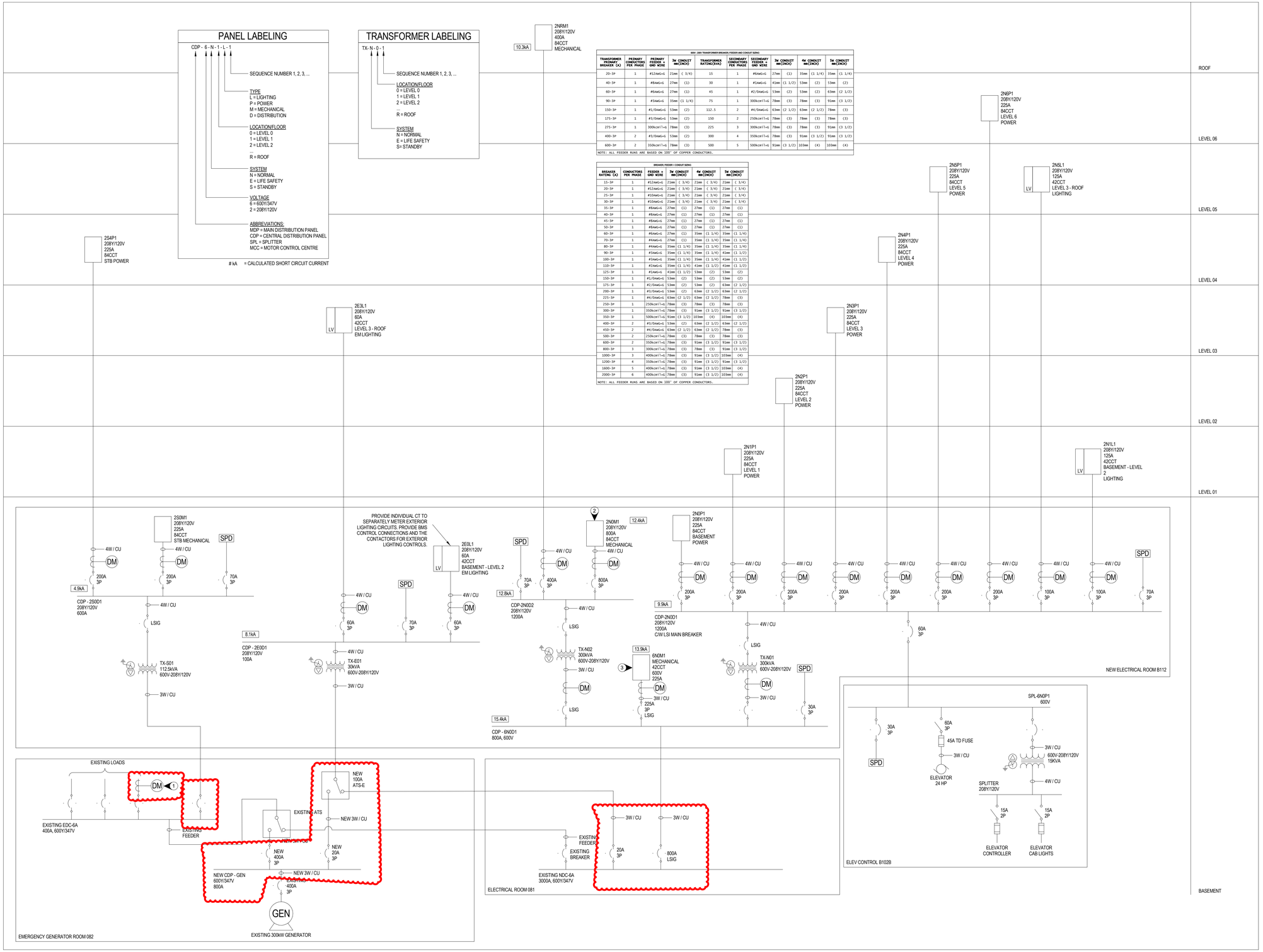
TRANSFORMER PRIMARY BREAKER (A)	PRIMARY CONDUCTORS PER PHASE	PRIMARY FEEDER + GND WIRE	3w CONDUIT mm(INCH)	TRANSFORMER RATING(KVA)	SECONDARY CONDUCTORS PER PHASE	SECONDARY FEEDER + GND WIRE	3w CONDUIT mm(INCH)	4w CONDUIT mm(INCH)	5w CONDUIT mm(INCH)
20-3P	1	#12AWG+G	21mm (3/4)	15	1	#6AWG+G	27mm (1)	35mm (1 1/4)	35mm (1 1/4)
40-3P	1	#8AWG+G	27mm (1)	30	1	#1AWG+G	41mm (1 1/2)	53mm (2)	53mm (2)
60-3P	1	#6AWG+G	27mm (1)	45	1	#2/0AWG+G	53mm (2)	53mm (2)	63mm (2 1/2)
90-3P	1	#3AWG+G	35mm (1 1/4)	75	1	300kcmil+G	78mm (3)	78mm (3)	91mm (3 1/2)
150-3P	1	#1/0AWG+G	53mm (2)	112.5	2	#4/0AWG+G	63mm (2 1/2)	78mm (3)	
175-3P	1	#3/0AWG+G	53mm (2)	150	2	250kcmil+G	78mm (3)	78mm (3)	78mm (3)
275-3P	1	300kcmil+G	78mm (3)	225	3	300kcmil+G	78mm (3)	78mm (3)	91mm (3 1/2)
400-3P	2	#3/0AWG+G	53mm (2)	300	4	350kcmil+G	78mm (3)	91mm (3 1/2)	91mm (3 1/2)
600-3P	2	350kcmil+G	78mm (3)	500	5	500kcmil+G	91mm (3 1/2)	103mm (4)	103mm (4)

NOTE: ALL FEEDER RUNS ARE BASED ON 100' OF COPPER CONDUCTORS.

BREAKER, FEEDER + CONDUIT SIZING

BREAKER RATING (A)	CONDUCTORS PER PHASE	FEEDER + GND WIRE	3w CONDUIT mm(INCH)	4w CONDUIT mm(INCH)	5w CONDUIT mm(INCH)
15-3P	1	#12AWG+G	21mm (3/4)	21mm (3/4)	21mm (3/4)
20-3P	1	#12AWG+G	21mm (3/4)	21mm (3/4)	21mm (3/4)
25-3P	1	#10AWG+G	21mm (3/4)	21mm (3/4)	21mm (3/4)
30-3P	1	#10AWG+G	21mm (3/4)	21mm (3/4)	21mm (3/4)
35-3P	1	#8AWG+G	27mm (1)	27mm (1)	27mm (1)
40-3P	1	#8AWG+G	27mm (1)	27mm (1)	27mm (1)
45-3P	1	#8AWG+G	27mm (1)	27mm (1)	27mm (1)
50-3P	1	#8AWG+G	27mm (1)	27mm (1)	27mm (1)
60-3P	1	#6AWG+G	27mm (1)	35mm (1 1/4)	35mm (1 1/4)
70-3P	1	#4AWG+G	27mm (1)	35mm (1 1/4)	35mm (1 1/4)
80-3P	1	#4AWG+G	35mm (1 1/4)	35mm (1 1/4)	35mm (1 1/4)
90-3P	1	#3AWG+G	35mm (1 1/4)	35mm (1 1/4)	41mm (1 1/2)
100-3P	1	#3AWG+G	35mm (1 1/4)	35mm (1 1/4)	41mm (1 1/2)
110-3P	1	#2AWG+G	35mm (1 1/4)	41mm (1 1/2)	41mm (1 1/2)
125-3P	1	#1AWG+G	41mm (1 1/2)	53mm (2)	53mm (2)
150-3P	1	#1/0AWG+G	53mm (2)	53mm (2)	53mm (2)
175-3P	1	#1/0AWG+G	53mm (2)	53mm (2)	63mm (2 1/2)
200-3P	1	#3/0AWG+G	53mm (2)	63mm (2 1/2)	63mm (2 1/2)
225-3P	1	#4/0AWG+G	63mm (2 1/2)	63mm (2 1/2)	78mm (3)
250-3P	1	250kcmil+G	78mm (3)	78mm (3)	78mm (3)
300-3P	1	350kcmil+G	78mm (3)	91mm (3 1/2)	91mm (3 1/2)
350-3P	1	500kcmil+G	91mm (3 1/2)	103mm (4)	103mm (4)
400-3P	2	#3/0AWG+G	53mm (2)	63mm (2 1/2)	63mm (2 1/2)
450-3P	2	#4/0AWG+G	63mm (2 1/2)	63mm (2 1/2)	78mm (3)
500-3P	2	250kcmil+G	78mm (3)	78mm (3)	78mm (3)
600-3P	2	350kcmil+G	78mm (3)	91mm (3 1/2)	91mm (3 1/2)
800-3P	3	300kcmil+G	78mm (3)	78mm (3)	91mm (3 1/2)
1000-3P	3	400kcmil+G	78mm (3)	91mm (3 1/2)	103mm (4)
1200-3P	4	350kcmil+G	78mm (3)	91mm (3 1/2)	91mm (3 1/2)
1600-3P	5	400kcmil+G	78mm (3)	91mm (3 1/2)	103mm (4)
2000-3P	6	400kcmil+G	78mm (3)	91mm (3 1/2)	103mm (4)

NOTE: ALL FEEDER RUNS ARE BASED ON 100' OF COPPER CONDUCTORS.



1 / E2.50 SINGLE LINE DIAGRAM
NTS

KEYNOTES	
1	PROVIDE METERING TRIACTA POWERHAWK FOR THE FOLLOWING EQUIPMENT: MCC-E1, MCC-E2, TR-E1, TR-E2, TR-E3.
2	PROVIDE SUBMETER TO MONITOR BRANCH CIRCUITS FOR THE FOLLOWING EQUIPMENT: DHWT-1, DHWT-2, DHWT-3. CONTRACTOR TO PROVIDE METERING SETRA SSP/SPM.
3	PROVIDE SUBMETER TO MONITOR BRANCH CIRCUITS FOR THE FOLLOWING EQUIPMENT: AHU-1, AHU-2, CH-1. CONTRACTOR TO PROVIDE METERING SETRA SSP/SPM.

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ISSUES + REVISIONS

NO	DATE	DESCRIPTION
1	2021-11-30	Issued for Schematic Design Submission
2	2022-09-15	Issued for 40% Construction Drawings
3	2022-10-25	Issued for Stage 1 BP
4	2022-10-25	Issued for Tender 1
5	2022-12-22	Issued for 75% CD
6	2023-01-12	Issued for Tender 2
7	2023-01-19	Issued for Full BP
8	2023-01-19	Issued for Stage 1 BP R1
10	2023-01-20	Post-Tender 2 - Addendum 02
11	2023-02-06	Post-Tender 2 - Addendum 03
15	2023-03-09	Issued for Construction 1
16	2023-04-19	Issued for Stage 2 BP
17	2023-04-21	IFC2 Progress Set
18	2023-05-02	Issued for IFP Electrical Package
19	2023-05-31	Issued for Construction 2

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PROJECT CODE
2-21-395
SCALE
1 : 1

Single Line Diagram

SHEET
E2.50